

JARED W. FULK

Jfulk44@gmail.com

linkedin.com/in/jared-f-9a14621b8 · github.com/jaredf96 · jareds.codes

Computer Science graduate (B.S., UNC Charlotte — Data Science specialization) who builds database-backed web applications, APIs, and workflow tools with React, TypeScript, Python/FastAPI, PostgreSQL, and AWS — focused on reliable data models, testing, and translating operational problems into maintainable software. A case study for every project below:

[jareds.codes/work](#).

PROJECTS

Health Analytics — Healthcare Data Warehouse in dbt (2026 · [github.com/jaredf96/health-analytics](#))

dbt · DuckDB · SQL · Dimensional modeling · Python · GitHub Actions

- Built a star schema over a synthetic EHR export: six staged feeds, six conformed dimensions, and an encounter fact at 61,459 rows, with billed, payer-covered, and patient-responsibility totals reconciling to the cent against source
- Wrote 180 data-quality tests (122 not_null, 20 unique, 14 relationships, 14 accepted_values, 10 singular) with every foreign key resolving to zero orphans; CI builds and publishes the generated docs on every push to main
- Applied HIPAA Safe Harbor de-identification in the mart, enforced by a test that reads information_schema so a reintroduced identifier fails the build; priced a known data defect at 0.27% of encounters rather than filtering it

Shut It Down — Cloud Lab Cleanup Dashboard (2026 · [github.com/jaredf96/shut-it-down-aws](#))

AWS · Python · FastAPI · React · DynamoDB · Boto3 · CloudFormation · Docker · Pytest + moto

- Scans 17 AWS regions concurrently (~12 s) for forgotten EC2, EBS, Elastic IP, NAT Gateway, load-balancer, RDS, and S3 resources — cross-account via STS assume-role with CloudFormation onboarding
- Estimates each finding's minimum monthly cost from AWS pricing data, explains in plain English why it bills and the safe next action, and alerts on new billable or riskier resources
- Guides opt-in cleanup behind seven safety gates — dry-run default, typed confirmation, live-state re-check, audit trail; 219 backend + 65 frontend tests in CI

CalTracker — Accuracy-First Calorie & Macro Tracker (2026)

SvelteKit · Svelte 5 · TypeScript · Supabase (Postgres, Auth, RLS) · Tailwind CSS · Vitest

- Per-100g canonical data model — every total is derived at read time, so a corrected food retroactively fixes past logs
- Four capture paths (camera barcode scanning via ZXing, label entry, USDA FoodData Central search, manual), all quarantined “needs review” until confirmed
- Postgres with row-level security via Supabase SSR auth; six unit-test suites over nutrition math, API clients, auth guard, and DB layer

Additional work — local-first gym tracker PWA, explainable resume/JD matching engine, Winston-Lutz medical QA dashboard, aviation-accident NLP analysis: [jareds.codes/work](#)

EXPERIENCE

FedEx Ground, Kernersville, NC — Package Handler (Nov 2024 – Mar 2025)

- Used barcode scanning and internal tracking systems to maintain accurate package flow in a high-volume logistics environment

EDUCATION

University of North Carolina at Charlotte — B.S. Computer Science, Data Science specialization (Aug 2020 – Dec 2023)

GPA 3.3 · Dean's List 2020 · Coursework: Software Engineering, Cloud Computing, Database Design & Implementation, Data Mining, Operating Systems & Networking

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++

Cloud & Backend: AWS (DynamoDB, Boto3, CloudFormation), FastAPI, Docker, Supabase, PostgreSQL (row-level security), Vercel

Frontend: React, SvelteKit/Svelte 5, Tailwind CSS, PWA / offline-first

Data & Analytics Engineering: dbt, DuckDB, dimensional modeling, data-quality testing, pandas, scikit-learn, NLP (TF-IDF, clustering), Matplotlib/Seaborn

Testing: Vitest, Pytest + moto, GitHub Actions CI

VOLUNTEERING

Tutoring (middle/high-school math, science, English) · American Red Cross · Second Harvest Food Bank